Power Spectral Density (PSD) Analysis of Phase Modulated (PM) Signals Using MATLAB

AIM :

- To generate a phase-modulated (PM) signal using MATLAB.
- To calculate and plot its frequency spectrum and power spectral density.
- To identify the carrier and sidebands and analyze the signal bandwidth.

CODE :

```
clc;

clear;

close all;


% Parameters

Am = 1;            % Message amplitude (V)

fm = 100;          % Message frequency (Hz)

Ac = 5;            % Carrier amplitude (V)

fc = 1000;         % Carrier frequency (Hz)

fs = 20*fc;        % Sampling frequency (Hz)

kp = pi/2;         % Phase sensitivity (rad/V)


% Time vector

t = 0:1/fs:4/fm;


% Message signal

m = Am*sin(2*pi*fm*t);
```

```
% Phase modulated signal
pm = Ac*cos(2*pi*fc*t + kp*m);

% Phase deviation
beta = kp*Am;

% FFT and frequency axis
N = length(pm);
f = linspace(-fs/2, fs/2, N);
PM_FFT = abs(fftshift(fft(pm)))/N;

% Power Spectral Density
PSD = (abs(fftshift(fft(pm))).^2)/(N*fs);

% Approximate bandwidth
BW = 2*(beta + 1)*fm;

% Display results
fprintf('Phase Deviation = %.2f radians\n', beta);
fprintf('PM Modulation Index = %.2f\n', beta);
fprintf('Estimated Bandwidth = %.2f Hz\n', BW);

% Plot signals and spectrum
figure('Name', 'PSD Analysis of PM Signal', 'NumberTitle', 'off');
```

```
subplot(2,2,1);
plot(t, m);
title('Message Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

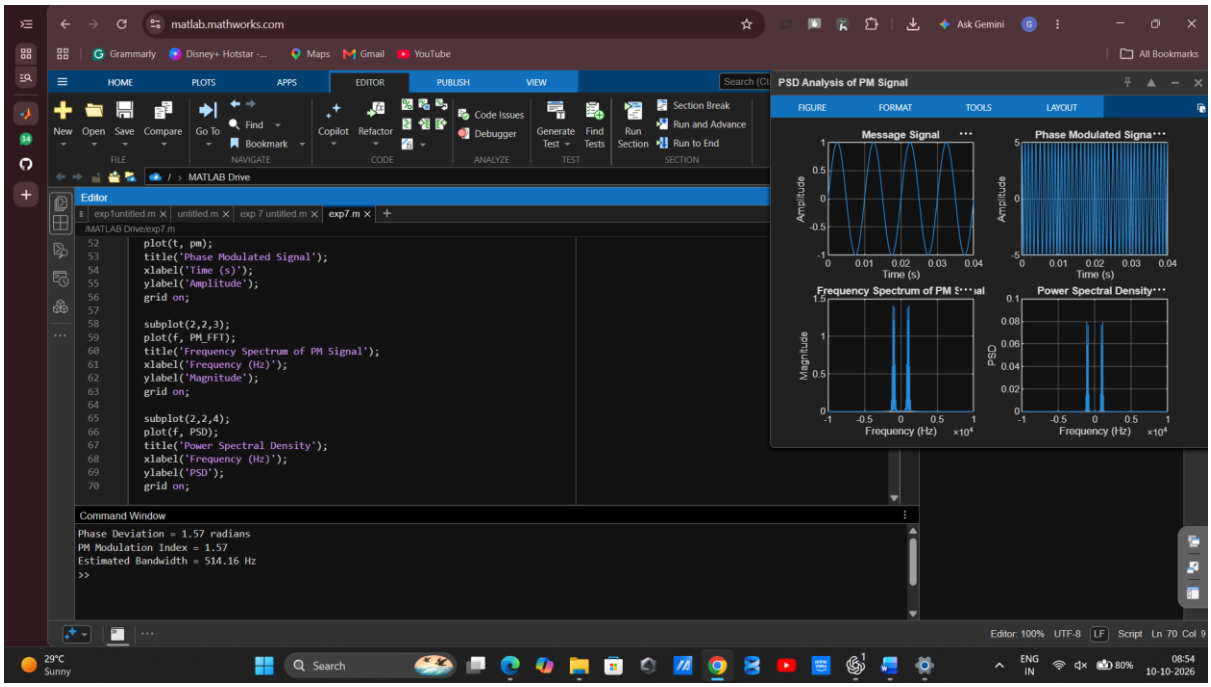
subplot(2,2,2);
plot(t, pm);
title('Phase Modulated Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

subplot(2,2,3);
plot(f, PM_FFT);
title('Frequency Spectrum of PM Signal');
xlabel('Frequency (Hz)');
ylabel('Magnitude');
grid on;

subplot(2,2,4);
plot(f, PSD);
title('Power Spectral Density');
```

```
xlabel('Frequency (Hz)');
ylabel('PSD');
grid on;
```

OUTPUT :



RESULT :

- The PSD of the PM signal was successfully calculated and plotted using MATLAB.
- The phase deviation was **1.57 radians**, and the estimated bandwidth was approximately **514 Hz**.
- The spectrum showed the carrier component and sidebands around the carrier frequency